Yousef Alaa Awad

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Engineering with Honors | GPA: 3.77 — Aug 2027
Relevant Courses: Verification of Digital Systems, HDL in Digital Systems Design, & Adv. Computer Architecture
Affiliations: IEEE-HKN (Zeta Chi Chapter), IEEE, ACM

Work Experience

Lockheed Martin Corporation — Orlando, FL
ASIC/FPGA Design CWEP Intern — Feb 2026 – Present

- Verified ASIC template by writing 20+ SystemVerilog Assertions (SVA) for bounded model checking in Synopsys VC Formal, capturing 3 critical corner-case bugs (overflows and race conditions) prior to dynamic simulation.
- Deployed a reproducible FPV sign-off suite within GitLab CI/CD utilizing parallel matrix jobs, standardizing formal verification workflows across 3 independent ASIC teams and eliminating manual regression bottlenecks.
- Modernized 8+ Synopsys VCS simulation workflows by reverse-engineering Make dependencies into modular Bash scripts, facilitating seamless cross-team adoption and establishing new technical standards for cross-module testbench referencing.

Embedded Software Engineering CWEP Intern — Apr 2025 – Feb 2026

- Designed a system-level Hardware-in-the-Loop (HIL) simulation environment in C++, enabling high-fidelity testing into already existing workflow for 20+ software engineers by implementing a CIGI v2 compliant image generator.
- Developed custom C++ automation tools for Hardware-in-the-Loop (HIL) test environments, reducing manual data analysis efforts by 20% and further optimized CI/CD pipeline via Bash scripts to decrease software integration time by 10%.
- Engineered the migration of legacy C++/Qt tooling environments from Visual Studio to a CMake build system, slashing compilation times by 50% and establishing a modular architecture to improve future scalability of the program.

Projects

KnightCore – Full RISC-V SoC – 2025 AMD Hardware Competition – *Group Project* — Apr 2025 – Aug 2025

- Developed a coverage-driven verification testbench in Python (Cocotb) to validate a custom 32-bit RISC-V SoC (RV32I ISA), implementing constrained-random stimulus to effectively explore the design state space.
- Developed a comprehensive test suite with directed tests for all ISA-defined instructions and constrained-random stimulus generation to stress the ALU and branch prediction logic, leading to the discovery and resolution of 5 critical RTL bugs.

2025 SouthEastCon Robotics Competition – Embedded Software Lead – *Group Project* — Sep 2024 – Apr 2025

- Led software pipeline for 10+ team using ROS2, enabling low-latency control between modular hardware and decision nodes. Used Gazebo and OpenCV for validation, contributing to 1st place in Design and 2nd in Performance at competition.
- Developed embedded firmware on Teensy 4.1 for sensor fusion and actuator control with modular, real-time C++ code. Optimized for high-frequency loops and low resource use, improving system responsiveness and performance.

CyndaQuil Compiler/Language – *Solo Project* — Sep 2024 – Jan 2025

- Designed and implemented a statically-typed compiled language (CyndaQuil) from scratch in C++, utilizing CMake to manage the build system across parsing, semantic analysis, and x86 assembly generation stages.
- Engineered low-level system components and optimized performance, integrating Assembly and C to enhance hardware-software interfacing, ensuring minimal latency and efficient computation for embedded systems.

Leadership

Eta Kappa Nu (IEEE-HKN)
Chapter President, Zeta Chi Chapter — Apr 2026 – Present

- Directing overall chapter operations and strategic vision, overseeing the executive board to ensure the continued growth and national recognition of the Zeta Chi Chapter.

Chapter Vice President, Zeta Chi Chapter — Apr 2025 – Apr 2026

- Lead the re-establishment of the dormant IEEE-HKN chapter, positioning the organization to achieve national *Key Chapter* status by growing membership to 31 students and developing a new recruitment pipeline.
- Architected the chapter's operational revival by engineering foundational processes for member induction and organizational scope, while simultaneously managing the financial and logistical strategies to double national conference delegation.

Institute of Electrical and Electronics Engineers
Treasurer, UCF Student Branch — Apr 2025 – Apr 2026

- Managed the IEEE UCF chapter's finances, developing and tracking over a $10,000+ budget across multiple projects and events, ensuring accurate allocation of resources and financial accountability.
- Facilitated over $6500 in sponsorship funding through effective communication with industry partners, directly supporting student project development and participation in competitions like the SouthEastCon Hardware Competition.

Technical Skills

Languages: SystemVerilog, Verilog, C++, Tcl, Python, C, Bash, Rust, Assembly (x86, RISC-V, MIPS)
Hardware Verification: Formal Property Verification (FPV), SystemVerilog Assertions (SVA), Functional Coverage
Computer Architecture: RISC-V ISA, AXI/Memory-Mapped Interfaces, Pipelining, Caching, Memory Hierarchy
Hardware & EDA Tools: Synopsys VC Formal, Synopsys VCS, Xilinx Vivado, Verilator
Systems & Scripting: Linux, Git, GitLab CI/CD, ROS/ROS2, Real-Time C++, CMake/Make